

Particle Physics — General Concepts

IMAPP lecture notes

Spring 2026

1 Natural units

Particle physics commonly uses $\hbar = c = 1$. The conversion factors are

$$\hbar = 1.055 \times 10^{-34} \text{ J s}, \quad c = 2.998 \times 10^8 \text{ m s}^{-1}, \quad 1 \text{ GeV} = 1.602 \times 10^{-10} \text{ J}.$$

In natural units, time has dimensions of inverse energy and length has dimensions of inverse energy:

$$[T] = E^{-1}, \quad [L] = E^{-1}.$$

Useful conversions are $1 \text{ GeV}^{-1} \simeq 0.197 \text{ fm}$ and $1 \text{ fm} \simeq 5.07 \text{ GeV}^{-1}$.

2 Mandelstam variables

For a $2 \rightarrow 2$ process with momenta $p_1, p_2 \rightarrow p_3, p_4$,

$$s = (p_1 + p_2)^2, \quad t = (p_1 - p_3)^2, \quad u = (p_1 - p_4)^2.$$

For fixed external masses, $s + t + u = \sum_i m_i^2$. In the centre-of-mass frame, $\sqrt{s}$ is the total energy.

3 Decay and the Breit-Wigner form

The population of unstable particles follows

$$N(t) = N(0)e^{-\Gamma t}, \quad \tau = \Gamma^{-1},$$

where Γ is the total decay width. A finite lifetime gives an energy distribution around the resonance energy E_0 :

$$\rho(E) \propto \frac{\Gamma/2}{(E - E_0)^2 + (\Gamma/2)^2}.$$

The full width at half maximum is Γ .

For a resonance R produced and then decaying, $A + B \rightarrow R \rightarrow A + B$, the cross-section is enhanced near $\sqrt{s} = M_R$. In the narrow-width approximation, $\Gamma \ll M_R$, the resonant factor approaches a delta function and production and decay can be approximately factorized:

$$\sigma(AB \rightarrow R \rightarrow X) \simeq \sigma(AB \rightarrow R) \text{Br}(R \rightarrow X).$$

Summary

Natural units simplify relativistic calculations, Mandelstam variables organize $2 \rightarrow 2$ kinematics, and unstable particles appear as Breit-Wigner resonances with width set by their lifetime.